ANTIGRAVITY IDE AGENT SPECIFICATION

TARGET APK: DURAGRIT-VENDOR-V4.8.2.APK (VENDOR & FIELD APP)

Vendor & Field Mobile APK — Surgical Implementation Blueprint

Zero-regression instructions for completing the missing hardware/file-handling features in PDF Steps 10 and 11 inside duragrit-vendor-v4.8.2.apk without altering existing screens (Steps 08, 09) or backend database tables.

STRICT NON-DESTRUCTIVE GUARDRAILS FOR ANTIGRAVITY IDE AGENT (READ FIRST)

- **DO NOT MODIFY** existing completed mobile screens: Step 08 (Vendor Sign In & Morbi GIDC Yard Selector), Step 09 (Contractor Dashboard, 18,450 Points Wallet & Profile Info), or the separate `duragrit-admin-command-v2.apk` (Step 12).
- **DO NOT ALTER BACKEND DATABASE TABLES OR API CONTRACTS:** All bulk CSV ingest items and proof attachments must map to the existing single-scan/claim submission payload structure or local device storage cache.
- **PRESERVE NAVIGATION & THEME:** Keep the bottom tab navigation (`Overview` , `Registry` , `Scan QR` , `Orders` , `Account`) and existing component hierarchy intact.

1. Scope of Work Matrix — Vendor & Field APK Only (Steps 10 & 11)

PDF Step	Target Screen & Zone	Exact Missing Feature in PDF	Non-Destructive Agent Action
Step 10A (Page 10)	Scan QR & Upload Camera Viewfinder Top Bar	`[ Torch Off / On ]` hardware flash toggle & live `DETECTING: DTA-TYPE3-HD` optical status pill.	Bind existing `Torch Off` button to camera track `torch: true/false` constraint (or Capacitor/RN flash API) with state toggle.
Step 10B (Page 10)	Scan QR & Upload Middle Card & Bottom Queue	1. `[ Switch to Manifest CSV / Photo Ingest ]` 2. Offline queue: `RECENT FIELD TAGS (SESSION) • 3 QUEUED`	Add CSV/Image file picker modal that parses serials (`DG-#####-###`) into the existing scan handler + offline `localStorage`/`AsyncStorage` sync queue.
Step 11A (Page 11)	My Cashback Requests Claim Card Bottom Row	`[ Invoice_DTA999_MOR.pdf ]` & `[ Scan_Batch_Receipt_08.jpg ]` with `View proof →` action link.	Add in-app `ProofAttachmentModal` to preview/upload PDF & JPG receipt proofs and bind `Manual Claim` button to attach proof files.

PDF Step	Target Screen & Zone	Exact Missing Feature in PDF	Non-Destructive Agent Action
Step 11B (Page 11)	My Cashback Requests Claim Card CB-9014 Row	`◎ Geo-tagged at Job Site #12` with `View details →` action link.	Add `ClaimGeoDetailsModal` displaying captured GPS coordinates, Dispatch Node (`Morbi Yard 04 Hub`), and NPCI escrow audit trail.

2. Task-by-Task Antigravity Agent Execution Schema

Task A: Step 10 — Camera Torch Hardware Toggle, Manifest CSV/Photo Ingest & Offline Session Queue

Context: On Page 10 (`← Scan Qr & Upload`), field contractors/vendors scan pallet and bag QR codes (`Format: DG-[5-digit Lot]-[Unit Identifier]`, e.g., `#DG-88410-B04`). Three functional bindings are required on the existing screen:

- **Step A.1 (Camera Torch Hardware Binding):** In the scanner component, connect the top-right `Torch Off` button to toggle hardware flash:
  - If WebView/Capacitor/PWA camera stream: call `videoTrack.applyConstraints({ advanced: [{ torch: !isTorchOn }] })`.
  - If React Native / Flutter camera controller: toggle `flashMode` (`torch` vs `off`) and update button label between `Torch Off` and `Torch On`.
- **Step A.2 (`Switch to Manifest CSV / Photo Ingest` Action):** Clicking `Switch to Manifest CSV / Photo Ingest` opens a lightweight bottom-sheet/modal (`ManifestIngestSheet`) with two tabs:
  - **Tab 1 — Manifest CSV Upload (`.csv` / `.txt`):** Parses uploaded CSV rows matching regex `/^#?DG-[A-Z0-9-]+$/i` (e.g., `DG-94812-A01`, `DG-94812-A02`) and sequentially submits each serial through the existing single-code submission handler (`Log Code`) so no new backend endpoint is required.
  - **Tab 2 — Photo Ingest (`.jpg` / `.png`):** Allows selecting a label photo from device gallery, decodes the QR/barcode client-side (or attaches photo to manual serial entry), and logs it into the session list.
- **Step A.3 (Offline Queue & Auto-Sync — `RECENT FIELD TAGS (SESSION)`):** Maintain a persistent local queue (`duragrit_field_tag_queue_v1`). When offline, newly scanned or CSV-imported tags appear with badge `QUEUED`; when connectivity is active, replay queued items against the existing scan API and transition badge to green `SYNCED` with timestamp (`14:22:08`).

Task B: Step 11 — Claim Proof Viewer (`View proof →`) & Geo-Tag Detail Modal (`View details →`)

Context: On Page 11 (`← My Cashback Requests`), each claim card (`CB-9021`, `CB-9018`, `CB-9014`, `CB-8995`) shows a bottom attachment/location bar (`Invoice_DTA999_MOR.pdf`, `Scan_Batch_Receipt_08.jpg`, `Geo-tagged at Job Site #12`) and a top `[ Manual Claim ]` button.

- **Step B.1 (`View proof →` In-App Document & Image Preview Modal):** Create a non-intrusive modal component (`ClaimProofPreviewModal`) triggered when the user taps `View proof →` or the attachment pill (`Invoice_DTA999_MOR.pdf`, `Scan_Batch_Receipt_08.jpg`, `TaxInv_9921_Dealer.pdf`):
  - Renders high-resolution image preview for `.jpg`/`.png` batch receipts with pinch-to-zoom.
  - Renders embedded invoice summary / PDF viewer for `.pdf` tax invoices with Claim ID (`CB-9021`), SKU (`DTA 999 • QR-88410-01`), Claim Value (`₹250`), Payout UPI (`rahulsharma@okaxis`), and a `Replace / Upload New Proof` button.
- **Step B.2 (`View details →` Geo-Tag & Escrow Audit Modal):** Clicking `View details →` on geo-tagged claims (`CB-9014` / `CB-8995`) opens `ClaimAuditDetailsModal` showing:
  - **Job-Site Geo-Tag:** Site label (`Job Site #12`), captured latitude/longitude (via device Geolocation API at scan time, stored locally/in existing metadata), and Dispatch Node (`Morbi Yard 04 Hub`).
  - **Verification Timeline:** `Submitted Timestamp (19 Oct 2026, 17:40)` → `DuraGrit Central Factory Serialization Ledger Cross-Audit (24–48 hrs)` → `NPCI Instant Clearing Payout (Sec 194C TDS)`.
- **Step B.3 (`[ Manual Claim ]` Proof Attachment Support):** Inside the existing `Manual Claim` flow on Step 11, allow the user to attach an optional Invoice PDF or Batch Receipt JPG (`max 5MB`) and capture current device GPS coordinates (`navigator.geolocation.getCurrentPosition`) without changing any required backend fields.

3. Copy-Paste Prompt for Antigravity IDE Agent

```
ROLE: Senior Mobile App Engineer (React Native / Capacitor / Hybrid APK) inside
Antigravity IDE.
TARGET BUILD: `duragrit-vendor-v4.8.2.apk` (Steps 08-11 Mobile Workflow).

STRICT CONSTRAINTS:
1. DO NOT modify Step 08 (Vendor Sign In), Step 09 (Contractor Dashboard), or
`duragrit-admin-command-v2.apk` (Step 12).
2. DO NOT alter backend database schemas or break existing API endpoints. Map all
bulk/proof operations to existing scan/claim state and local storage.

EXECUTE ONLY THESE 2 SCREENS:
SCREEN 1 (Step 10 - Scan QR & Upload):
- Bind `Torch Off` button in the camera header to toggle hardware camera
flash/torch (`Torch On` / `Torch Off`).
- Wire `Switch to Manifest CSV / Photo Ingest` to open `ManifestIngestSheet`
supporting bulk `.csv` serial parsing (`DG-#####-###`) and gallery label image QR
decoding, feeding each code into the existing `Log Code` submission handler.
- Add offline-resilient session queueing (`duragrit_field_tag_queue_v1`) for
`RECENT FIELD TAGS (SESSION)` so offline scans queue and auto-sync to `SYNCED`.

SCREEN 2 (Step 11 - My Cashback Requests):
- Wire `View proof ->` on claim cards (`Invoice_DTA999_MOR.pdf`,
`Scan_Batch_Receipt_08.jpg`) to open `ClaimProofPreviewModal` displaying the
invoice/receipt preview.
- Wire `View details ->` on `Geo-tagged at Job Site #12` cards to open
`ClaimAuditDetailsModal` showing GPS coordinates, Dispatch Node (`Morbi Yard 04
Hub`), and NPCI verification stage.
- Allow optional PDF/JPG receipt attachment and auto GPS capture inside the `Manual
Claim` action.
```